Physiology

Departmental Objectives

At the end of the course in physiology the **MBBS** students will be able to:

- Demonstrate basic knowledge on the normal functions of human body and apply it as a background for clinical subjects.
- Explain normal reactions to environment and homeostatic mechanism.
- Interpret normal function with a view to differentiate from abnormal function.
- Demonstrate knowledge & skill for performing and interpreting physiological experiments.
- Develop knowledge and skill to proceed to higher studies and research in physiology in relation to need and disease profile of the country.
- Develop sound attitude for continuing self-education to improve efficiency & skill in physiology.

List of Competencies to acquire :

Medical courses in physiology teach the essentials of the processes of life.

The physiology courses are very clinically relevant because the knowledge of the processes underlying the normal physiological functions of all the major organ systems is crucial for understanding pathology, pharmacology, and for competent clinical practice. In fact, all of medicine is based on understanding physiological functions.

In the process of completing these courses, students acquire the following competencies:

- Describe transport across the plasma membrane, the basis of resting membrane potential, the genesis and propagation of action potentials. Explain muscle excitation and contraction.
- Describe the heart and circulation and how the circulatory system functions as a dual pump and dual circulatory system with the knowledge of properties of cardiac muscle, cardiac cycle, hemodynamics, heart rate and blood pressure.
- Explain respiratory processes with the knowledge of structures, ventilation, diffusion, blood flow, gas transport, mechanics of breathing, and control of ventilation.
- Identify how the kidney plays an important role in the maintenance of homeostasis by regulating both the composition and volume of ECF compartment.
- Explain how the brain works at the neuronal systems level. The role of electrical & chemical signals in information transmission & processing. Brain circulation, metabolism, neurotransmitter release & receptors,
- Describe the physiological mechanism underlying sensory perception, motor control & maintenance of homeostasis as well as higher cortical functions. Understanding autonomic nervous system.
- Describe endocrine physiology: describe the synthesis, secretion, functions & mechanism of action of the endocrine hormones.
- Explain human reproduction, functional changes in the reproductive tract, the formation of sperm & ovum, fertilization & hormonal regulation of fertility, role of hormones in pregnancy, parturition & lactation.
- The students will be able to equip themselves with adequate knowledge and develop skill for performing physiology laboratory tests and interpreting these normal functions with a view to differentiate from abnormal conditions. such as
- Measurement of blood pressure
- Examination of radial pulse.
- Recording & analysis of normal ECG (electrocardiogram)(12 Lead).
- Auscultation of heart sounds, breath sounds & bowel sound.
- Estimation of Hb concentration.
- Estimation of total count of red blood cell (RBC).
- Estimation of total and differential count of white blood cell **(**WBC).
- Determination of bleeding time & clotting time.
- Determination of blood grouping & cross matching.
- Determination of erythrocyte sedimentation rate (ESR).

- Determination of packed cell volume.
- Measurement of pulmonary volumes & capacities.
- Examination of urine for volume, specific gravity/osmolarity and water diuresis.
- Elicitation of reflexes (e.g., knee jerk, ankle jerk, planter response, biceps jerk, triceps jerk).
- Recording of body temperature.
- Elicitation of light reflex.
- Interpretation of Snellen's chart and colour vision chart.
- Conduction and interpretation of Rinne test.
- Conduction and interpretation of Weber test.

Organization of the Course:
The course is offered in 3 terms (1st, 2nd & 3rd) total one & half years for phase -I MBBS Course.

Distribution of teaching - learning hours

Lecture	Tutorial	Practical	Total Teaching hours	Integrated teaching for Phase I	Formative Exam		Summative exam	
					Preparatory leave	Exam time	Preparatory leave	Exam time
120 hrs	120 hrs	97 hrs	337 hrs	36hrs	35 days	42 days	30days	30 days
Time for integrated teaching, examination, preparatory leave of formative & summative assessment is common for all subjects of the phase								
Related behavioral, professional & ethical issues will be discussed in all teaching learning sessions								

Teaching/learning methods, teaching aids and evaluation

Teaching Methods			Teaching aids	In course evaluation
Large group	Small group	Self learning		
Lecture Integrated teaching	Tutorial Practical Demonstration	Assignment, self assessment & self study.	Computer & Multimedia & other IT materials Chalk & board White board & markers OHP Slide projector Flip Chart Models Specimens projector Study guide & manuals.	Item examination (oral) Practical item examination (Oral & practical) Card completion examination (written) Term final Examination (Written, oral & practical)

1st Professional Examination:
Marks distribution of Assessment of Physiology
Total marks – 400 (Summative)

- Written= 200 (SAQ + SEQ) 140 + MCQ (SBA+MTF) 40+Formative 20)
- SOE =100
- Practical= 100 (OSPE40 + Traditional 50 +Note Book 10)

Related Equipments:
Microscope, test tube, glass slide, centrifuge machine, micro pipette, chemicals & reagents, Sphygmomanometer, Stethoscope, ECG machine, Spirometer, Peak flow meter, Urinometer, clinical hammer, cotton, pin, clinical thermometer, spirit, pencil torch, Ishihara charts, Snellen's chart, tuning fork, models, specimens, Haemocytometer, Shahlis haemometer, haematocrit tube, westergren ESR tube & ESR stand etc.

Learning Objectives and Course Contents in Physiology

Cellular Physiology

Learning Objectives	Contents	Hours / days
At the end of the course the students will be able to: • explain goal of physiology. • explain principles of homeostasis • describe functional organization of the human body & cell physiology. • describe cell membrane transport. • Explain membrane potential, resting membrane potential and action potential. • describe muscle physiology • describe neuromuscular junction.	**CORE:** • Physiology: definition, goal & importance of physiology. • Homeostasis: definition, major functional systems, control systems and regulation of the body function. • The cell: functions of cell membrane and cell organelles. • The cell membrane transport: active & passive transport, exocytosis & endocytosis, intercellular communication, • Membrane potential: definition, basic physics of membrane potential. Resting membrane potential. • Action potential: definition & propagation of action potential. • Mechanism of skeletal muscle contraction & relaxation. • Neuromuscular junction: transmission of impulse from nerve ending to muscle fibre.	L=5 T=6 P=2

Physiology of Blood

Learning Objectives	Contents	Hours / days
At the end of the course the students will be able to: • describe the composition & functions of blood. • demonstrate knowledge about plasma proteins. • demonstrate knowledge about the formation , morphology, types & functions of RBC,WBC & platelets. • describe synthesis & breakdown of haemoglobin. • demonstrate knowledge about the blood grouping & blood transfusion. • describe about hemostasis & coagulation. • describe about the bleeding disorders.	**CORE:** • Blood: composition & functions. • Plasma proteins: origin, normal values, properties, functions & effect of hypoproteinaemia • Development and normal values of formed elements. • RBC: erythropoiesis. • Hemoglobin: synthesis, types, functions & fate of hemoglobin. • Red blood cell indices, • Anaemia, Polycythemia & Jaundice: definition & classification. • WBC: Classification, morphology, properties & functions, leucocytosis, leucopenia. • Platelet: morphology & functions. • Hemostasis: definition & events. • Coagulation: definition, mechanism, • Clotting factors & fibrinolysis • Blood grouping: ABO & Rh system • Hazards of blood transfusion & Rh incompatibility. **Additional/Applied Physiology** • Bleeding disorder: thrombocytopenic purpura & hemophilia, tests for bleeding disorder .	L=15 T=16 P=45 IT=06

Cardiovascular Physiology

Learning Objectives	Contents	Hours / days
At the end of the course the students will be able to : • describe the physiology of cardiac muscle • describe the rhythmical excitation of the heart. • demonstrate knowledge about events of cardiac cycle. • explain about the heart sounds. • explain about a normal ECG. • describe about hemodynamics. • describe local & humoral control of blood flow by the tissues. • describe the microcirculation, capillary fluid & interstitial fluid • describe about cardiodynamics: cardiac output, venous return & peripheral resistance. • explain about the heart rate & radial pulse. • describe the regulation of blood pressure. • demonstrate knowledge about the coronary circulation. • demonstrate knowledge about shock • describe the circulatory changes during exercise.	**CORE :** • Cardiac muscle: physiological anatomy, properties. • Junctional tissues of the heart: generation of cardiac impulse & its conduction. • Cardiac cycle: events, pressure & volume changes during different phases • Heart sounds: types & characteristics • ECG: principles, characteristics & interpretations • Functional classification of blood vessels & microcirculation • Interrelationship among pressure, flow & resistance. • Local & humoral control of blood flow by the tissue. • Exchange of fluid through the capillary membrane. • SV, EDV, ESV, EF: definition & factors affecting them. • Cardiac output: definition, measurement, regulation and factors affecting cardiac output. • Venous return: definition & factors affecting. • Peripheral resistance: definition & factors affecting. • Heart rate: definition, normal values, factors affecting & regulation. • Radial pulse: definition & characteristics. • Blood pressure: definition, types, measurement & regulation of arterial blood pressure. **Additional /Applied Physiology** Circulatory adjustment during exercise. Coronary circulation Cardiac arrhythmias: tachycardia, bradycardia & heart block Shock: definition, classification. Physiological basis of compensatory mechanism of circulatory shock.	L=18 T=18 P=18 IT=03

Respiratory Physiology

Learning Objectives	Contents	Hours / days
At the end of the course the students will be able to : • define pulmonary & alveolar ventilation. • explain the mechanism of respiration • describe pulmonary volumes and capacities, • describe pulmonary circulation • explain the diffusion of gases through the respiratory membrane. • describe the oxygen & carbon dioxide transport. • describe the respiratory centers & regulation of respiration. • define & classify hypoxia and cyanosis.	**CORE** • Physiological anatomy of respiratory system • Respiration: definition, mechanism. • Pulmonary & Alveolar ventilation. • Pulmonary volumes and capacities (spirometry) • Dead space: definition & types • Pulmonary circulation- pressure in pulmonary system effect of hydrostatic pressure in lungs, pulmonary capillary dynamics. • Composition of atmospheric, alveolar, inspired and expired air. • Respiratory unit and respiratory membrane. • Diffusion of Gases through the respiratory membrane. • Transport of Oxygen & Carbon dioxide in blood & body fluid. Oxy-hemoglobin dissociation curve. Bohr effect, Haldane effect & chloride shift mechanism. • Respiratory centers: name, location & functions. • Nervous & chemical regulation of respiration. • Lung function tests: name, significance • Ventilation -perfusion ratio. • Regulation of respiration during exercise. • Hypoxia: definition, types • Cyanosis: definition & types. **Additional/Applied Physiology** • Oxygen therapy in hypoxia • Definition of dyspnea, hypercapnea & periodic breathing.	L=12 T=14 P=8 IT=03

Renal Physiology

Learning Objectives	Contents	Hours / days
At the end of the course the students will be able to: • describe the structure & function of nephron. • describe the mechanism of urine formation. GFR, tubular reabsorption, tubular secretion. • describe the mechanism of water balance and osmotic diuresis. • explain physiological mechanism of micturition.	**CORE:** • Kidney: functions • Nephron: types, parts, structure & functions • Renal circulation: peculiarities & functional importance • Urine formation: basic mechanism • GFR: definition, determinants, measurement, control of GFR & regulation of renal blood flow • Reabsorption and secretion by the renal tubules • Definition of $T_{m,}$ Renal threshold, tubular load & plasma load, plasma clearance and diuresis, • Mechanism of formation of concentrated urine & diluted urine. • Micturition reflex **Additional /Applied Physiology** Abnormalities of micturition	**L= 12** **T= 10** **P= 02** **IT=06**

Gastrointestinal Physiology

Learning Objectives	Contents	Hours / days
Gastrointestinal Physiology At the end of the course the students will be able to: • describe the general principles of gastrointestinal function. • describe the movements of GIT	**CORE:** • Phygiological anatomy of gastrointestinal (GI) tract. • Enteric nervous system. • Local hormones of GIT: name, function & regulation of secretion • Hormonal control of GI function. • Movements of the GIT. • GI reflexes. • Functions of stomach, small intestine and large intestine **Additional / Applied Physiology** Pyloric pump	L=10 **T=8** P=02 IT=03

Endocrine Physiology and Physiology of Reproduction

Learning Objectives	Contents	Hours / days
At the end of the course the students will be able to: • describe types, hormonal receptors & general mechanism of action of hormone. • describe functions, mechanism of action & regulation of secretion of individual hormone. • describe disorders in relation to pituitary gland, thyroid and parathyroid gland, adrenal gland and endocrine pancreas	**CORE :** • Endocrine glands : name & name of their hormones. • Hormone: definition, classification, mechanism of action, assessment of hormone level. • Hypothalamic hormones, releasing & inhibitory hormones: name and functions. • Pituitaty Gland: physiological anatomy. • Pituitary hormones (anterior & posterior): name, functions, mechanism of action and their control by the hypothalamus and disorders (dwarfism, gigantism, acromegaly & hypopituitarism and diabetes insipidus). • Thyroid Gland: physiological anatomy. • Thyroid hormones: biosynthesis, transport, functions, mechanism of action, regulation of secretion, disorders (hypo and hyperthyroidism, cretinism, myxoedema and goitre).. • Parathyroid Gland: physiological anatomy. • Parathyroid hormone: functions, mechanism of action & regulation of secretion. • Adrenal Gland: physiological anatomy. Adrenocortical hormones: name, functions , mechanism of action , regulation of secretion & disorders (Addison's disease, Cushing's Syndrome, Conn's disease). • Islets of Langerhan's of pancreas - hormones: functions, mechanism of action & regulation of secretion **Additional / Applied Physiology** Pathophysiology of insulin deficiency.	L=20 T=20 P=02 IT=06

<table>
<tr><th>Learning Objectives</th><th>Contents</th><th>Hours / days</th></tr>
<tr><td>Physiology of Reproduction

At the end of the course the students will be able to :
• describe male & female reproductive organs & their hormones
• describe spermatogenesis
• explain about functions of testosterone, oestrogen and progesterone
• describe ovulation, ovarian & menstrual cycle
• demonstrate knowledge about puberty
• explain about lactation</td><td>• Introduction to reproductive physiology, sex determination & sex differentiation. Puberty
• Functional anatomy of male reproductive system
• Secondary sex characteristics of male
• Testes: functional structure and functions
• Testosterone: function.
• Spermatogenesis: steps & hormonal control.
• Functional anatomy of female reproductive system
• Secondary sex characteristics of female
• Ovaries : functional structure and functions. Functional structure of uterus.
• Menstrual cycle: definition, phases and hormonal control.
• Ovarian cycle: phases and hormonal regulation.
• Ovulation: definition, mechanism & hormonal control.
• Definition of menstruation, menarche & menopause.
• Ovarian hormones
• Functions of oestrogen and progesterone.
• Placental hormones: name & functions.
• Mammogenesis: development and lactation.

Additional/Applied Physiology
Indicators of ovulation.
Anovulatory cycle.</td><td></td></tr>
</table>

Neurophysiology

Learning Objectives	Contents	Hours / days
At the end of the course the **Students will be able to:** • explain organization of the nervous system • explain the basic mechanism of synaptic transmission. • describe the sensory system of the body. • describe the organization and functions of the spinal cord. • explain the spinal cord reflexes. • describe the motor control system- pyramidal and extra pyramidal systems. • describe the functions of cerebellum. • describe functions of basal ganglia, thalamus, reticular formation & limbic system • describe functions of hypothalamus • describe organization & function of autonomic nervous system	**CORE:** • Functional organization of nervous system and functions of major levels of central nervous system(CNS). • **Neuron:** definition, parts, types • **Nerve fiber:** classification, properties, effects of injury/section to the nerve fiber • **Synapse:** physiological anatomy, properties, types, synaptic transmission • **Neurotransmitters:** definition, types, functions • **Sensory receptor:** definition, classification, properties, receptor potential. • **General/somatic senses:** definition, classification • **Ascending tracts/sensory pathways:** name & function. • Spinothalamic tract, tract of Gall, tract of Burdach, spinocerebellar tract : origin, course, termination & function. • **Cerebral cortex:** name & functions of the Brodmann's areas • **Reflex:** definition, classification, properties, • **Reflex arc:** definition, components • Stretch reflex, withdrawal reflex, crossed extensor reflex, reciprocal innervation & planter response. • **Muscle spindle**: definition, physiological anatomy, functions. • **Muscle tone**: definition, function, maintenance • **Descending tracts / motor pathways:** name & function. • Pyramidal tract: origin, course, termination, function & effect of lesion. • Extrapyramidal tract: name, functions. • Upper motor neuron and Lower motor neuron: definition, example, effect of lesion. • Spinal cord: hemisection.	L=18 T=18 P=08 IT=03

Learning Objectives	Contents	Hours / days
	• **Cerebellum:** functional division, functions, error control mechanism of motor activity & cerebellar disorder. • **Basal ganglia:** functional components, functions & effects of lesion • **Thalamus, Reticular formation, limbic system:** components & functions. • **Hypothalamus:** name of the nucleus and functions • **Autonomic Nervous system:** components and functions **Additional/Applied Physiology** Pain: types, dual pathway for transmission of pain, referred Pain. Thermostatic function of hypothalamus. Posture, equilibrium: definition, name of the areas controlling them. Sleep, memory: definition, name of the areas controlling them. Alarm or stress response.	

Physiology of Body Temperature

<table>
<tr><th>Learning Objectives</th><th>Contents</th><th>Hours / days</th></tr>
<tr><td>At the end of the course the students will be able to:
describe the physiology & regulation of body temperature.</td><td>CORE :
Normal body temperature, site of measurement, sources of heat gain, channels of heat loss, regulation of body temperature in hot and cold environment.
Additional/Applied Physiology
Heat stroke, hypothermia, frost bite, fever.</td><td>L=02
T=02
P=02</td></tr>
<tr><td colspan="3">Physiology of Special Senses</td></tr>
<tr><td>At the end of the course the students will be able to:
describe the neurophysiology of vision and visual pathway
explain errors of refraction, accommodation reaction, light reflexes, dark and light adaptation.
explain mechanism of hearing and describe auditory pathway
describe the physiology of smell and taste</td><td>CORE:
Vision : physiological anatomy of eye, image formation in the eyes, visual receptors, visual pathway, common refractive errors, photochemistry of vision, accommodation reaction, light reflex , dark & light adaptation, Field of vision, color vision, color blindness, visual acuity.
Hearing: auditory apparatus, receptor, mechanism of sound wave transmission, auditory pathway.
Smell: smell receptors, olfactory pathway.
Taste: taste receptors, modalities of taste sensation, taste pathway.
Additional/Applied Physiology
Effects of lesion in visual pathway.
Argyll Robertson pupil, Horner’s syndrome.</td><td>L=08
T=08
P=08
IT=06</td></tr>
</table>

Physiology Practical

Learning Objectives	Contents	Hours / days
Cellular Physiology & Physiology of Blood **Students will be able to** • demonstrate knowledge on common laboratory equipments used for practical hematology. • perform common hematological tests. • interpret results for practical purpose.	**CORE:** • Developing skill in using of microscope & common laboratory equipments. • Collection & preparation of blood sample. • Observation of osmotic behavior of RBC • Determination of total count of RBC, • Determination of total count of WBC • Determination of differential count of WBC. • Estimation of haemoglobin. • Observation of osmotic fragility of RBC. • Determination of ESR • Determination of PCV. • Determination of Blood grouping (ABO & Rh system) & cross matching. • Determination of bleeding time & clotting time. • Interpretation of Red Cell Indices	02 45
Cardiovascular Physiology **Students will be able to :** • examine the radial pulse & its application. • measure the blood pressure and effect of exercise on it. • auscultate 1^{st} & 2^{nd} heart sounds. • record & analysis of normal ECG.	**CORE :** • Measurement of Blood Pressure & effect of exercise on it. • Auscultation of 1^{st} & 2^{nd} heart sounds. • Examination of radial pulse. • Recording & analysis of normal ECG (12 leads).	**18**

Learning Objectives	Contents	Hours / days
Respiratory Physiology **Students will be able to :** • examine the Respiratory system • perform lung function tests & interpret tests on clinical conditions. • demonstrate the knowledge about breath sounds.	**CORE:** • Examination of respiratory system (physiological aspect) • Counting of respiratory rate. • Auscultation of breath sounds. • Determination of lung function tests including Spirometry.	08
Gastrointestinal Physiology Students will be able to: • auscultate the intestinal sound	**CORE** • Auscultation of intestinal sound	02
Renal Physiology Students will be able to: • Determine the specific gravity of urine	**CORE** • Determination of specific gravity of urine	02
Neurophysiology Students will be able to : • examine the sensory & motor functions of human body. • elicit the reflexes & interpret its clinical importance.	**CORE :** • Examination of motor & sensory functions. • Elicitation of the reflexes & interpretation of its clinical importance. (knee jerk, biceps jerk, triceps jerks & planter response).	10
Physiology of Body Temperature Students will be able to • record the body temperature	**CORE:** • Recording of the body temperature. • Observation of the effect of exercise on body temperature.	**02**
Physiology of Special senses Students will be able to : • perform the light reflex & accommodation reaction • perform visual acuity & color vision. • conduct tests for hearing & interpret the result	**CORE:** • Observation of Light reflex, • Interpretation of visual acuity and color vision. • Conduction and interpretation of Rinne test & Weber test.	08

Distribution of Teaching Hours

Systems	Lecture hours	Tutorial hours	Practical hours	Integrated teaching hours
1. Cellular Physiology	5	6	2	
2. Physiology of blood	15	16	45	6
3. Cardiovascular Physiology	18	18	18	3
4. Respiratory Physiology	12	14	8	3
5. Gastrointestinal Physiology	10	8	2	3
6. Renal physiology	12	10	2	6
7. Endocrine Physiology & Physiology of Reproduction	20	20	2	6
8. Neurophysiology & Physiology of body temperature	20	20	10	3
9. Physiology of Special Senses	08	8	8	6
Total	120	120	97	36

Time allocation in Physiology in different term

Term	Lecture hours	Tutorial hours	Practical hours	Integrated teaching hours	Total hours
1st Term	38	40	35	03	116
2nd Term	34	32	32	04	98
3rd Term	48	48	30	03	126
Grand Total	120	120	97	10	337

Summative Assessment of Physiology
(First Professional Examination)

Assessment systems and mark distribution

<table>
<tr><th>Components</th><th>Marks</th><th>Total Marks</th><th>Contents</th></tr>
<tr><td>WRITTEN EXAMINATION
Paper – I- Formative Assessment + MCQ +SAQ
Paper – II- Formative Assessment + MCQ +SAQ</td><td>10+20+70 = 100
10+20+70 =100</td><td>200</td><td rowspan="4"><u>Paper – I</u>
1. Cellular physiology
2. Physiology of blood
3. Cardiovascular physiology
4. Respiratory physiology
5. Gastrointestinal physiology

<u>Paper – II</u>
1. Renal physiology
2. Endocrine physiology & physiology of Reproduction
3. Neurophysiology & temperature regulation
4. Physiology of Special senses</td></tr>
<tr><td>PRACTICAL EXAMINATION
OSPE
Traditional practical methods and experiments
Practical Note Book</td><td>40
50
10</td><td>100</td></tr>
<tr><td>ORAL EXAMINATION
(Structured)
2 boards</td><td>Board – I = 50
Board – II = 50</td><td>100</td></tr>
<tr><td colspan="2">Grand Total</td><td>400</td></tr>
</table>

Pass marks 60% in each of written, oral and practical.

Academic Calendar for Physiology

		1st Term		**2nd Term**		**3rd Term**	
Teaching /Learning Method	**Teaching hours including Examination**	20 Working weeks	**E** **V** **A**	20 Working weeks	**E** **V** **A**	18 Working weeks	**E** **V** **A**
Lecture	120 Hours	GP- 05 hours Blood—15 hours CVS—18 hours	**L** **U** **A**	Resp. Physiology—12 hours GIT—10 hours Renal- 12 hours.	**L** **U** **A**	Endocrine & Reproduction—20 hours Nervous system & Body temp.—20 hours. Special Senses-08 hours.	**L** **U** **A** **T**
Tutorial	120 hours	GP—06 hours. Blood –16 hours. CVS—18 hours.	**T** **I** **O** **N**	Respiration—14 hours. GIT—08 hours. Renal —10hours.	**T** **I** **O** **N**	Endocrine & reproduction—20 hours. Nervous system & Body temp. –20 hours Special Senses—08 hours.	**I** **O** **N**
Practical	100 hours.	GP—02 hours. Blood—36 hours.	**4** **W** **E** **E** **K** **S**	Blood-- 12 hours CVS---18 hours. GIT—02 hours	**4** **W** **E** **E** **K** **S**	Respiration- 08 hours Renal – 02 hours Endocrine—02 hours Neuro physiology -08 hours Body temp—**02** hours Special Senses--08 hours	**7** **W** **E** **E** **K** **S**